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Assignment No. 01
Algorithmic Assignment
Due Date: 18-05-2024 10:00 PM

Q#1 :

Following are programs expressed as English statements. Write only the programming

statements for the actual program and their output.

The variable x starts with the value 10

The variable y starts with the value 2

The variable z starts with the value 4

Store the value of x times y in x

Store the value of y times z in y

Add x and z and store the result in y

What is the value of y if the output on screen?

SOLUTION :

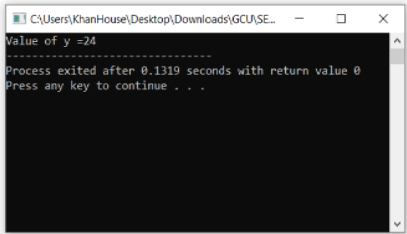
The value of y will be : **24**

OUTPUT :

Value of y=24

↓ The picture is of the written form
of the code given below on dev.

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     int x=10;
6     int y=2;
7     int z=4;
8     x=y*x;           // x=2*10=20
9     y=z*y;           // y=4*2=8
10    y=x+z;           // y=20+4
11    cout<<"Value of y ="<<y; //value of y = 24
12 }
13
```



Q#2 :

Following algorithm has an error. The program is supposed to ask the user for the length

and width of a rectangular room, and then display the room's area. The program must

multiply the width by length in order to determine the area.

area = width into length

Display "What is Room's Width"

Input width

Display "What is Room's Length"

Input length

Display area

SOLUTION :

The error occurs because the formula for calculating the area is declared before the variables for length and width are declared. As a result, the variables used in the area calculation (length and width) are unknown at the time of processing, leading to an inability to compute the area.

**Error is shown
in the code
written on the
compiler**



```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     int area;
6     area=length*width;
7     int length=5,width=5;
8     cout<<area;
9
10 }
11
```

Line 6: [Error] 'length' was not declared in this scope
Line 6: [Error] 'width' was not declared in this scope

Q#3 :

Which of the following is a Character Literal

'A'

"A"

SOLUTION :

'A' is the Character Literal.

Q#4 :

Assuming the size of data types required in memory is 1B,4B,8B for Character, integer

and double respectively. How many bytes the following declaration required

int x,y,z;

char a,b;

double j,k;

SOLUTION :

Given Data:

Integer Size = 4B

Character Size = 1B

Double Size = 8B

Required:

Total memory size =?

Solve:

- For the **integers x, y, and z**, each requires 4 bytes, so that's $3 \times 4 = 12$ bytes.
- For the **characters a and b**, each requires 1 byte, so that's $2 \times 1 = 2$ bytes.
- For the **doubles j and k**, each requires 8 bytes, so that's $2 \times 8 = 16$ bytes.

Adding them all together:

12 bytes+2 bytes+16 bytes=**30 bytes**

So, the total memory required for the given declarations is 30 bytes.

Q#5 :

How would you consolidate the following definitions into one statement.

```
int x = 7;
```

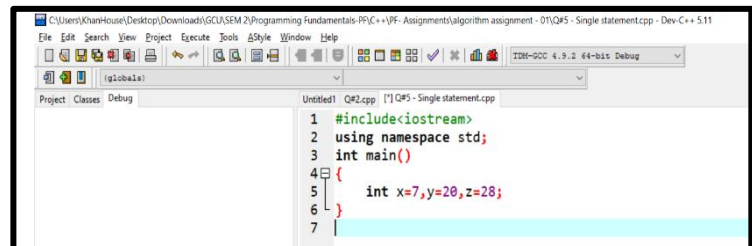
```
int y = 20;
```

```
int z = 28;
```

SOLUTION :

To write the given values in single statement we use comma operator such as:

```
int x=7,y=20,z=28;
```



Q#6 :

How many operands does each of the following types of operator require?

Unary , Binary and Ternary

SOLUTION :

- Unary operators required only **one** operator. Examples of unary operators are Increment Operator (++), Decrement Operator (--), Logical operation (!)
- Binary operators require only **two** operators. Examples of binary operators are Addition (+), Subtraction (-), Multiplication (*), Division (/), Modulus (%), Assignment (=), Logical AND (&&), Logical OR (||), Bitwise AND (&)
- Ternary operators required only **three** operators. Examples of ternary operators conditional operator.

Q#7 :

How the group of statements are enclosed to treat as a block.

SOLUTION :

In programming languages, the group of statements is enclosed within a block using curly brackets.

For example in C++

```

if (condition)
{
statement1;
statement2;
// more statements
}

```



Representation of how a group of statements is treated as a block

Q#8:

Assume $w = 5$, $x = 4$, $y = 8$, and $z = 2$. What value will be stored in result in each of the following statements?

```

result = x + y ;
result = z * 2 ;
result = y / x ;
result = y - z ;
result = w % 2 ;

```

SOLUTION :

```

result = x + y =12
result = z * 2 =4
result = y / x =2
result = y - z =6
result = w % 2 =6

```

Q#9:

Write expressions for the following algebraic expressions.

$$y = 6x$$

$$a = 2b + 4c$$

$$y = x^2$$

$$g = \frac{x+2}{z^2}$$

$$y = \frac{x^2}{z^2}$$

SOLUTION :

Following are the algebraic expressions;

$$y=6*x;$$

$$a=2*b+4*c;$$

$$y=x*x;$$

$$g=(x+2)/z*z;$$

$$y=(x*x)/(z*z);$$

Q#10:

Design an algorithm to find the perimeter and area of rectangle. Formula used to find the

solution are

$$\text{Perimeter} = 2 * (\text{length} + \text{width})$$

$$\text{Area} = \text{length} * \text{width}$$

SOLUTION :

The algorithm to find the perimeter and area of the rectangle are as follows

Initialize the variable of L,W, P,A for length, width, perimeter, and area

Display "Enter the value of length:"

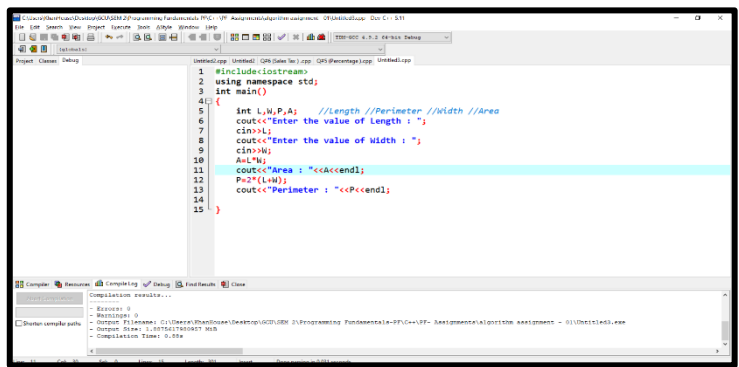
Input length;

Display "Enter the value of width:"

Input width;

Area=length into width

Display area;
Perimeter=2 into(length+width)
Display Perimeter



```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     int L,W,P,A; //Length //Perimeter //Width //Area
6     cout<<"Enter the value of Length : ";
7     cin>>L;
8     cout<<"Enter the value of Width : ";
9     cin>>W;
10    A=L*W;
11    cout<<"Area : "<<A<<endl;
12    P=2*(L+W);
13    cout<<"Perimeter : "<<P<<endl;
14
15 }
```

Q#11:

Design an algorithm to calculate the sales tax and price of an item sold in a particular state. The sales tax is calculated as follows. The state's portion of sales tax is 4% and the city portion of sales tax is 1.5%. if cost of the item is more than 50000 then a luxury tax of 10% is applicable.

SOLUTION :

- Initialize variable as Sales_Tax, Cost
 - Display "Enter cost of the item :"
 - Input Cost
 - Display " Enter the state :"
 - Input state
 - Display "Enter City"
 - Input city
- ★ **CONSTANTS:**
- ★ State's portion of sales tax rate: 4%
 - ★ City's portion of sales tax rate: 1.5%
 - ★ Luxury tax rate: 10%
 - ★ Luxury tax threshold: \$50,000
- Multiply the item cost by the sum of state and city sales tax rates to get the base sales tax.
 - If the item cost is above the luxury tax threshold, calculate the luxury tax as 10% of the item cost.
 - Add the base sales tax and the luxury tax (if applicable) to get the total sales tax.
 - Add the cost of the item and the total sales tax to get the total price.
 - Display the "total sales tax and the total price."

Q#12:

Given the radius, in inches, and price of pizza, design an algorithm to find the price of pizza per square inch.

SOLUTION :

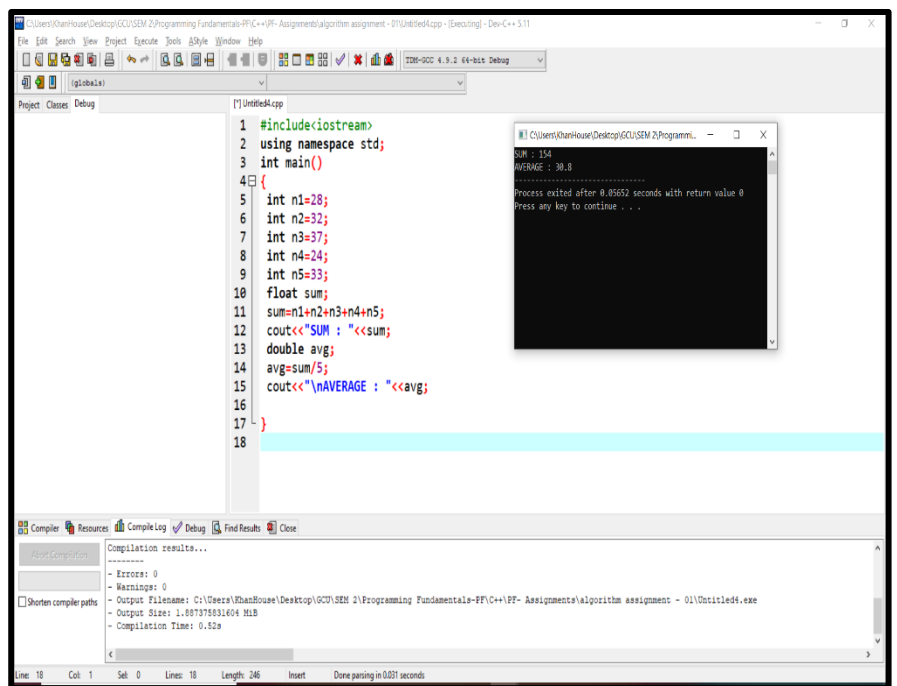
- Display “ Enter the radius of the pizza (in inches)”
 - Input radius
 - Display “ Enter the price of the pizza”
 - Input price_of_pizza
 - **Calculate** the Area of the pizza
 - $\text{Area_of_pizza} = \text{PI} * r * r$
 - **Calculate** the Price per Square Inch
 - Divide the price of the pizza by the area of the pizza to find the price per square inch:
 - $\text{Price_per_square_inch} = \text{Price_of_pizza} / \text{Area}$
 - Display “Price per inch”
-

Q#13:

To get the average of a series of values, you add the values up and then divide the sum by the number of values. Write a complete program that stores the following values in five different variables: 28 , 32 , 37 , 24 and 33. The program should first calculate the sum of these five number and then stores then in a different variable named sum. Then the program should display the average on screen. [Use data types correct]

SOLUTION :

```
#include<iostream>
using namespace std;
int main()
{
    int n1=28;
    int n2=32;
    int n3=37;
    int n4=24;
    int n5=33;
    float sum;
    sum=n1+n2+n3+n4+n5;
```



The screenshot shows a C++ IDE with the following code in the main editor:

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     int n1=28;
6     int n2=32;
7     int n3=37;
8     int n4=24;
9     int n5=33;
10    float sum;
11    sum=n1+n2+n3+n4+n5;
12    cout<<"SUM : "<<sum;
13    double avg;
14    avg=sum/5;
15    cout<<"\nAVERAGE : "<<avg;
16 }
17
18
```

The output window on the right shows the following output:

```
SUM : 154
AVERAGE : 30.8
Process exited after 0.05652 seconds with return value 0
Press any key to continue . . .
```

The bottom panel shows the compilation results:

```
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\KhanHouse\Desktop\SEM 2\Programming Fundamentals-PF\C++\Assignments\algorithm assignment - 01\Untitled4.exe
- Output Size: 1.087975531404 KB
- Compilation Time: 0.52s
```



```
cout<<"SUM : "<<sum;
double avg;
avg=sum/5;
cout<<"\nAVERAGE : "<<avg;
return 0;
}
```
